

Towards Anatomically Plausible Human Image Generation via Synthetic Localized Preferences

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[britton-li.github.io/ASAP](https://github.com/britton-li)

Abstract. Large-scale text-to-image foundation models have achieved remarkable visual realism, yet generating human images with correct anatomical structures remains challenging. Existing approaches enforce anatomical constraints through part-specific modules or localized loss weighting during supervised fine-tuning on high-quality human photos, but such datasets are limited and often provide ambiguous optimization signals due to confounding factors such as lighting, pose, and background. Preference-based alignment offers an alternative, but standard Direct Preference Optimization (DPO) treats all pixels equally and therefore fails to exploit the localized nature of anatomical artifacts. To address this, we propose the framework of Alignment via Synthetic Anatomical Preference (ASAP), which constructs controlled preference pairs through a localized degradation mechanism applied to high-fidelity human images. This mechanism performs a controlled experiment on images by introducing explicit anatomical errors in targeted regions while preserving the remaining content. With this mechanism, we create the Human Anatomical Preference (HAP) dataset with over 10K curated pairs for effective anatomical alignment of text-to-image human image generative models. To better leverage the locality of these controlled preference pairs, we introduce a localized and margin-bounded variant of DPO that prioritizes optimization in targeted anatomical regions while enforcing a finite preference margin to prevent over-optimization and preserve global semantics. We further introduce HAF-Bench, a benchmark for systematic evaluation of anatomical fidelity. Extensive experiments demonstrate that ASAP consistently reduces anatomical errors across multiple foundation models while maintaining overall image quality.

Keywords: Human Image Generation · Anatomical Plausibility · Preference Optimization · Alignment

1 Introduction

Text-to-image foundation models [4, 14, 26, 40] have made remarkable progress in achieving high visual realism. However, generating human images with correct

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Fig. 1: Effectiveness of ASAP. While foundation models like FLUX and SDXL frequently struggle with complex human anatomy (FLUX/SDXL-Base), especially for fingers, ASAP significantly improves anatomical plausibility (FLUX/SDXL-ASAP).

anatomical structures remains a challenge [11, 12, 15, 18, 48]: existing foundation models frequently produce severe anatomical artifacts, such as malformed hands with incorrect digit counts or distorted facial features (Fig. 1). As humans are highly sensitive to anatomical details in faces, hands, and poses, these artifacts significantly degrade perceived quality and create an uncanny valley effect.

To address this, previous approaches have attempted to enforce localized anatomical constraints by designing part-specific modules [37, 48] or applying localized loss weighting [15, 39] on high-fidelity human photos during supervised fine-tuning (SFT). Nevertheless, the availability of high-quality human image datasets is limited, and the optimization signals from these datasets are often ambiguous due to the presence of multiple confounding factors (*e.g.*, lighting, pose, background). Recent works such as HG-DPO [23] leverage preference datasets—where each data point consists of a pair of images, with a high-quality real photo preferred over a generated one—to align the learned model with human preferences, typically with Direct Preference Optimization (DPO) [27, 35]. However, standard DPO treats all pixels equally, increasing the likelihood of every pixel in the preferred image without leveraging local information about which regions contain anatomical errors.

In light of this limitation, we propose **Alignment via Synthetic Anatomical Preference (ASAP)**, a framework comprising two key components: (1) **Synthetic Preference Pairs**, which constructs anatomically degraded nega-

tives through a localized degradation mechanism on high-fidelity human images, and (2) **Anatomical DPO**, a localized and margin-bounded variant of DPO.

The localized degradation mechanism can be understood as a controlled experiment on images: we selectively corrupt specific regions (*e.g.*, hands) using existing generative models to create negative samples where the degradation is explicit and directly attributable, yielding the **Human Anatomical Preference (HAP) dataset**, comprising over 10K curated preference pairs specifically designed to target anatomical artifacts prevalent in human image generation.

To effectively leverage these preference pairs, Anatomical DPO introduces a localized weighting mechanism that explicitly prioritizes optimization signals within targeted anatomical regions. However, localization alone is insufficient, as standard unbounded optimization inevitably leads to over-optimization of these focused regions. To address this, we reformulate alignment as a margin-bounded regression problem. By enforcing a finite target margin, this objective prevents over-optimization while maintaining global semantic fidelity.

Moreover, since no existing benchmark comprehensively evaluates human image generation with anatomically plausible structures, we introduce the **Human Anatomical Fidelity Benchmark (HAF-Bench)**. HAF-Bench comprises 500 carefully curated prompts spanning five anatomical categories, *i.e.*, Basic Anatomy, Semantic Gestures, Self-Interaction, Object Interaction, and Full Body, along with a structured evaluation protocol and comprehensive metrics designed to rigorously assess fine-grained anatomical correctness.

Our main contributions are as follows:

- **Synthetic Anatomical Preferences:** We construct controlled preference pairs via a localized degradation mechanism that introduces anatomical errors in targeted regions while preserving global composition.
- **Anatomical DPO:** We propose a localized and margin-bounded DPO variant that prevents over-optimization while maintaining the generative prior.
- **Anatomical Benchmark:** We establish novel HAF-Bench to evaluate anatomical correctness. Extensive experiments demonstrate that ASAP significantly reduces anatomical errors across foundation models (18.4%↓ for FLUX, 11.8%↓ for SDXL, Anatomical Error Rate) while preserving visual quality (80.6% for FLUX, 88.5% for SDXL, Human Preference Voting).

2 Related Work

2.1 Human Image Generation

Human image generation remains a long-standing challenge in image synthesis due to the inherent structural complexity of the human body and the heightened perceptual sensitivity of humans to anatomical errors [11]. While modern foundation models [4, 14, 26, 30, 40] based on diffusion [7, 33, 34] and flow matching [17, 19] have achieved impressive visual realism, they continue to produce anatomical artifacts on highly articulated regions, such as incorrect finger counts and distorted facial structures.

To mitigate these issues, several works introduce mechanisms that explicitly focus on anatomically sensitive regions during supervised fine-tuning. Some approaches design part-aware architectures, for example by training specialized modules for hands or faces and integrating them with mixture-of-experts frameworks [37, 48]. Other methods apply localized supervision signals, such as mask-weighted losses or body-part-specific constraints, to encourage correct structural generation [15, 39]. While these strategies strengthen guidance toward anatomically plausible structures, they primarily rely on positive supervision from curated human image datasets. As a result, they lack explicit optimization signals that directly penalize anatomical errors, which limits their ability to reliably correct such artifacts during generation. Alternatively, recent methods like HumanRefiner [5] and RealisHuman [36] rely on multi-stage inpainting to correct anatomical errors. While effective for local details, this decoupled pipeline significantly increases inference time and risks compromising global image coherence.

2.2 Alignment with Human Preference.

As foundation models are typically trained with huge and diverse unlabeled datasets [6, 25, 28], the sample distributions they capture generally deviates from what humans prefer. In image generation, such misalignment is often observed as inferior aesthetics [31], text-image misalignment [13, 21, 41, 44], image artifacts [16], etc. This misalignment issue is much more severe when it comes to human image generation since humans can easily identify visual artifacts of human faces and bodies in generated images [22]. To align pretrained models with human preferences, reinforcement learning with human feedback (RLHF) [24, 47] has been widely adopted, where a reward model trained on human preference data guides reinforcement learning. A simpler alternative, direct preference optimization (DPO) [27], directly finetunes the model using preference pairs by increasing the likelihood of preferred samples and decreasing that of disfavored ones. Originally proposed for large language models, DPO has been adapted to diffusion and flow-matching models [35, 49], which are among the most popular visual generative models. However, DPO adjusts the likelihood of all pixels in an image uniformly, which can be inefficient since human preferences often depend on only a small subset of regions. Recent works [10, 38, 43] therefore introduce region-based alignment strategies to correct localized issues such as incorrect object composition or style mismatches. Nevertheless, these approaches have not yet been validated on the challenging problem of anatomical plausibility in human image generation.

2.3 Preference Data Synthesis and Curation.

No matter alignment is done through RLHF or DPO, a preference dataset is required to capture human preferences. For alignment of visual generative models, many large-scale preference datasets were constructed [13, 21, 41, 42, 44] in the sense that the data distribution differ by image semantics, textual description granularity, image style, geography/demography, annotation quality and many

other aspects. However, most existing datasets provide only *image-level* preferences, where annotators choose the better sample among candidates that may differ in multiple aspects simultaneously, such as aesthetics, prompt adherence, background composition, and localized artifacts. This makes the supervision signal inherently ambiguous, as it is unclear which factors explain the preference. Several works therefore explore improved data curation strategies, such as collecting ratings or rankings in addition to pairwise comparisons to train more reliable reward models for text-to-image evaluation [21, 44]. Other approaches attempt to reduce confounding factors during pair construction by enforcing stronger visual consistency between compared images [9]. However, these methods still rely on preferences emerging from naturally generated samples, where the source of preference remains implicit. In contrast, our ASAP framework synthesizes preference pairs through controlled localized degradations, ensuring that the preference difference can be directly attributed to specific anatomical errors.

3 Preliminaries

Diffusion Models and Flow Matching Models. The prevalent image generation models are mostly built with diffusion models (*e.g.*, Stable Diffusion [30]) and flow matching models (*e.g.*, FLUX [14]). Given that a Gaussian probability path is commonly used in flow matching models, the sampling process can be unified as sampling $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and simulating $d\mathbf{x} = v(\mathbf{x}, t)dt + g(t)dw$ to obtain $\mathbf{x}_1$ ³, where $v(\mathbf{x}, t)$ is the velocity field, dw is Brownian motion (*i.e.* infinitesimal Gaussian noise) and $g(t)$ is the standard deviation of this process. The model is trained with the following loss function:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[w(t) \|v_\theta(\mathbf{x}_t, t) - u(\mathbf{x}_1, \mathbf{x}_0, t)\|^2 \right]. \quad (1)$$

where $u(\mathbf{x}_1, \mathbf{x}_0, t)$ is the target velocity field induced by the reference path and $w(t)$ is a weighting factor depending on the model design (often set to 1). For instance, in diffusion models, $u(\mathbf{x}_t, \mathbf{x}_0, t) = \dot{\alpha}_t \mathbf{x}_0 + \dot{s}_t(\mathbf{x}_t - \alpha_t \mathbf{x}_0)/s_t$, where $\alpha(t)$ and $s(t)$ are the diffusion schedules; in a popular variant called rectified flow [19], $u(\mathbf{x}_1, \mathbf{x}_0, t) = \mathbf{x}_1 - \mathbf{x}_0$.

Direct Preference Optimization. Direct Preference Optimization (DPO) [27] aligns generative models with human preferences directly using preference datasets without any reinforcement learning process. For diffusion models, Diffusion-DPO [35] has successfully adapted this framework by utilizing the model training loss as a proxy for log-likelihoods and building an approximate DPO loss. Specifically, given a condition $\mathbf{c}$ and a preference pair $(\mathbf{x}_w, \mathbf{x}_l)$, the approximate preference gap is defined as [49]:

$$\delta_\theta(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c}) \approx -\left(\mathcal{L}^\theta(\mathbf{x}_w) - \mathcal{L}^\theta(\mathbf{x}_l) - (\mathcal{L}^{\text{ref}}(\mathbf{x}_w) - \mathcal{L}^{\text{ref}}(\mathbf{x}_l)) \right), \quad (2)$$

³ This is in the opposite direction of the time arrow in the convention of diffusion models.

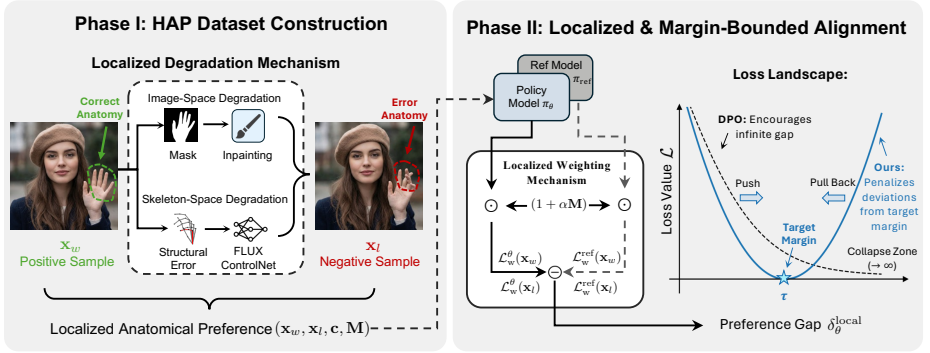


Fig. 2: Overview of the ASAP framework. Phase I (Left): Construction of the HAP dataset. We employ a localized degradation mechanism via image-space inpainting and skeleton-space perturbations to synthesize locally degraded negatives ($\mathbf{x}_l$). These negatives differ from the positives ($\mathbf{x}_w$) exclusively in anatomical correctness. **Phase II (Right):** Localized and Margin-Bounded Alignment Objective. To prioritize optimization signals, we introduce a spatial weight map ($\mathbf{M}$) to scale the preference signals via $(1 + \alpha \mathbf{M})$. Unlike standard DPO that drives the preference gap towards infinity ($\delta_\theta \rightarrow \infty$) causing global semantic collapse, we reframe alignment as bounded margin regression. By enforcing a target margin (τ), it effectively penalizes over-optimization, rectifying local anatomical structures while preserving the global image quality.

where $\mathcal{L}$ implicitly conditions on $\mathbf{c}$ for brevity. Under the assumption of Bradley-Terry preference model [2, 27, 35], the so-called Diffusion-DPO loss is

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c}) \sim \mathcal{D}} [\log \sigma(\beta \delta_\theta(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c}))], \quad (3)$$

where σ is the sigmoid function and β implicitly controls the magnitude of deviation from the reference model. A recent work shows that flow matching models can be similarly finetuned with it [49] to yield satisfactory results.

4 Methods

4.1 Synthesis of Localized Anatomical Preference Pairs

Positive Sample Generation. We first establish a high-quality anchor set $\mathcal{D}_{pos}$ with accurate anatomical structures and their corresponding human images. Instead of using existing datasets in the domain of human image generation, we adopt a pose-conditioned generation pipeline to create synthetic images for this dataset, so as to avoid the domain gap between photorealistic images and synthetic images during preference learning. Specifically, we extract precise human pose skeletons $\mathbf{p}$ via DWPose [46] and refine the corresponding textual prompts into detailed image caption $\mathbf{c}_{aug}$ using a Vision-Language Model (VLM) [3]. These paired conditions guide a pretrained Flux-ControlNet [32] to synthesize candidate positive images $\mathbf{x}_w \sim \mathcal{G}(\mathbf{c}_{aug}, \mathbf{p})$. The dataset is later examined by human annotators to filter out anatomically implausible samples.

Localized Degradation Mechanism. For each positive sample in $\mathcal{D}_{pos}$, we perform localized degradation so that the visual quality only differ at the designated regions, as illustrated in Fig. 2. Specifically, we consider two ways of degradation: 1) *image-space degradation* and 2) *skeleton-space degradation*. In *image-space degradation*, we use GroundingSAM2 [29] to extract precise binary regional masks $\mathbf{M}$ for highly articulated regions such as hands and faces, and only these masked regions are re-generated with an inpainting model conditioned on negative prompts that give explicit instructions of local distortions. In *skeleton-space degradation*, we perturb the pose condition $\mathbf{p}$ in the positive set by artificially corrupting the skeletal connections (*e.g.*, via synthesizing polydactyly or erroneous connectivity) to obtain the corrupted pose set $\mathbf{p}_{error}$. This corrupted condition, alongside the dense description $\mathbf{c}_{aug}$, is fed to FLUX-ControlNet [32] to regenerate $\mathbf{x}_l$.

HAP Dataset Construction. With the approaches described above with degradation mechanism randomly chosen, we curate a large candidate pool of preference pairs comprising about 25K raw preference pairs. By filtering out invalid pairs, we create Human Anatomical Preference (HAP) dataset. Specifically, HAP comprises over 10K highly controlled preference pairs $(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c})$. During the filtering stage, we prune invalid data based on three strict criteria: (1) high-fidelity positive samples, (2) clearly visible anatomical degradations in the negative samples, and (3) strictly preserved background compositions across each pair. To quantitatively validate these criteria, we characterize the final dataset using region-wise metrics. Specifically, the targeted anatomical regions exhibit significant structural divergence (average SSIM of 0.5315 and LPIPS of 0.3446), whereas the non-target areas remain exceptionally consistent (average SSIM of 0.9827 and LPIPS of 0.0109). These statistics confirm that our synthetic pipeline provides a controlled, and meaningful preference signal.



Fig. 3: Visual examples from the HAP dataset. Each column represents a preference pair conditioned on a specific text prompt. High-fidelity positive samples ($\mathbf{x}_w$) are shown in the top row, with correct anatomy highlighted in green boxes. Synthesized degraded negative samples ($\mathbf{x}_l$) are in the bottom row, with anatomical artifacts highlighted in red boxes.

4.2 Localized and Margin-Bounded Alignment

Both to further enforce the locality constraint and to avoid overfitting during DPO finetuning on the HAP dataset, we adopt a modified DPO algorithm that computes margin-constrained losses with segmentation masks.

Localized Preference Gap. To prioritize the optimization on the target anatomical regions (*e.g.*, hands and faces), we compute the individual loss terms in the DPO loss (Eq. (3)) by multiplying the difference between predicted and ground-truth velocities with the segmentation mask M and a positive scalar α :

$$\mathcal{L}_w(\theta; \mathbf{x}, \mathbf{c}, \mathbf{M}) = \mathbb{E}_{t, \mathbf{x}_0} \left[\|(1 + \alpha \mathbf{M}) \odot [v_\theta(\mathbf{x}_t, t, \mathbf{c}) - (\mathbf{x}_1 - \mathbf{x}_0)]\|^2 \right]. \quad (4)$$

The localized preference gap is thus ($\mathcal{L}_w^\theta(\mathbf{x})$ stands for $\mathcal{L}_w(\theta; \mathbf{x}, \mathbf{c})$):

$$\delta_\theta^{\text{local}}(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c}, \mathbf{M}) \approx - \left(\mathcal{L}_w^\theta(\mathbf{x}_w) - \mathcal{L}_w^\theta(\mathbf{x}_l) - (\mathcal{L}_w^{\text{ref}}(\mathbf{x}_w) - \mathcal{L}_w^{\text{ref}}(\mathbf{x}_l)) \right). \quad (5)$$

Margin-Bounded Regression. Since DPO is known to be prone to overfitting [20, 37, 45] and the size of the HAP dataset is relatively small, we adopt the approach of margin-constrained DPO [1] to the DPO variant for flow matching models (Eq. (3)) with our localized preference gap:

$$\mathcal{L}_{\text{Bounded}}(\theta) = \mathbb{E}_{(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c}, \mathbf{M}) \sim \mathcal{D}} \left[(\delta_\theta^{\text{local}}(\mathbf{x}_w, \mathbf{x}_l, \mathbf{c}, \mathbf{M}) - \tau)^2 \right], \quad (6)$$

where τ is the margin that regularizes the localized preference gap.

5 HAF-Bench

To rigorously evaluate the capability of generative models in handling complex human structures, we introduce HAF-Bench (Human Anatomical Fidelity Benchmark), a specialized evaluation suite curated to comprehensively assess fine-grained anatomical plausibility. Unlike standard benchmarks that prioritize general aesthetics, HAF-Bench specifically focuses on the anatomical plausibility of hands and faces.

Dataset Curation and Taxonomy.

We curated a diverse evaluation set comprising 500 challenging prompts designed to capture the long-tail distribution of anatomical anomalies. To ensure both diversity and difficulty, we employed an LLM-assisted generation pipeline guided by our hierarchical taxonomy, followed by rigorous manual curation to remove ambiguous or physically implausible descriptions. The benchmark is balanced across five distinct anatomical categories (100 prompts each), probing specific generative capabilities: (1) *Basic Anatomy*—correct body proportions and structural integrity, (2) *Semantic Gestures*—accurate hand poses conveying meaning,

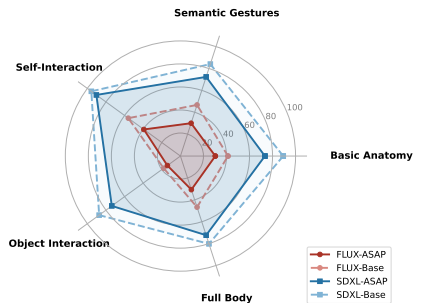


Fig. 4: AER Evaluation on HAF-Bench. Category-wise AER ($\downarrow$) across five anatomical domains. The smaller polygons for ASAP-tuned models demonstrate consistent AER improvements across all anatomical categories.

(3) *Self-Interaction*—realistic self-contact scenarios, (4) *Object Interaction*—plausible hand-object manipulation, and (5) *Full Body*—anatomical coherence across the entire figure.

Evaluation Protocols. Evaluating fine-grained anatomical plausibility at scale requires metrics closely aligned with human perception. We establish a standardized evaluation protocol using a state-of-the-art Vision Language Model (*i.e.*, Gemini-3-Pro-Preview), as a visual judge to detect anatomical anomalies. We employ two complementary metrics that assess generation quality from both absolute and relative perspectives.

Anatomical Error Rate (AER). To measure absolute fidelity, we define the Anatomical Error Rate (AER) to quantify the prevalence of anatomical hallucinations. The VLM serves as a binary detector to identify anatomical artifacts. AER is computed as the fraction of generated samples in the benchmark $\mathcal{D}_{\text{HAF}}$ that fail the anatomical assessment:

$$\text{AER} = \frac{1}{|\mathcal{D}_{\text{HAF}}|} \sum_{\mathbf{x} \in \mathcal{D}_{\text{HAF}}} \mathbb{I}(\mathcal{S}_{\text{VLM}}(\mathbf{x}) = \text{Fail}), \quad (7)$$

where $\mathcal{S}_{\text{VLM}}$ denotes the scoring function and $\mathbb{I}$ is the indicator function. Lower AER indicates higher anatomical plausibility.

Relative Superiority Index (RSI). To benchmark comparative performance, we propose the RSI metric. Standard win-rates are highly sensitive to the frequency of tie cases, often resulting in unstable performance estimations. To provide a more robust evaluation, RSI mitigates this instability by contrasting the frequency of our model’s non-inferiority (wins and ties) against the baseline’s non-superiority (losses and ties). Specifically, with the VLM acting as a preference oracle to output a decision $d \in \{\text{Win}, \text{Tie}, \text{Lose}\}$, the RSI is formulated as:

$$\text{RSI} = \frac{N_{\text{win}} + N_{\text{tie}}}{N_{\text{lose}} + N_{\text{tie}}}, \quad (8)$$

where N denotes the count of decisions. An RSI greater than 1 indicates that our model is more likely to match or exceed the baseline’s quality than to be outperformed by it. This metric proves to be reliable with an observed agreement rate of 82% between the VLM judgment (Gemini-3-Pro) and the human evaluation on the entire HAF-Bench dataset.

6 Experiments

6.1 Experimental Settings

Base Models. We consider two base models to experiment with: FLUX.1-dev [14] (a flow matching model) and SDXL [26] (a diffusion model). We note that FLUX model contains more number of parameters, is trained with better data and generate images of better quality compared to SDXL.

Table 1: Quantitative Comparison on HAF-Bench. We evaluate performance across two generative backbones: FLUX and SDXL. For anatomical assessment, AER measures anatomical errors (lower is better), while RSI indicates the preference win rate against the baseline (higher is better). Additionally, standard preference metrics (HPSv3, HPS v2.1, PickScore, ImageReward, AesScore) are reported to evaluate global visual aesthetics and text-image alignment. Notably, our proposed ASAP achieves the best anatomical plausibility without compromising the global image quality.

Models	AER ↓	RSI ↑	HPSv3 ↑	HPSv2 ↑	PickScore ↑	ImageReward ↑	AesScore ↑
FLUX-Base	0.417	1.00 (Ref.)	10.307	0.303	22.797	1.037	5.793
FLUX-SFT	0.391	1.03	10.320	0.303	22.808	1.097	5.822
FLUX-DPO	0.293	1.13	9.68	0.299	22.476	1.062	5.752
FLUX-ASAP (Ours)	0.289	1.17	10.360	0.304	22.814	1.102	5.779
SDXL-Base	0.873	1.00 (Ref.)	6.371	0.277	22.205	0.737	5.709
SDXL-SFT	0.820	1.23	7.233	0.289	22.369	0.770	5.766
SDXL-DPO	0.775	1.49	6.526	0.279	22.254	0.751	5.723
SDXL-ASAP (Ours)	0.764	1.58	7.450	0.291	22.364	0.778	5.754

Baselines. We consider two baseline methods: (1) Supervised Fine-Tuning (SFT), which directly finetunes the model on the correct anatomical samples from our dataset (*i.e.*, positive-sample-only finetuning); (2) vanilla DPO (without our localized preference gap and margin-based design).

Evaluation Benchmarks and Metrics. As explained in our evaluation protocol, we consider AER and RSI as our major metrics. In the meantime, to quantize the compromise in instruction following and visual quality due to our anatomical alignment we also report standard metrics for text-image alignment and/or image aesthetics, including HPSv3 [21], HPSv2 [42], PickScore [13], ImageReward [44] and Aesthetic Score [31].

Implementation Details. For parameter-efficient fine-tuning across both generative backbones, we use LoRA [8] with a rank of 32. Both models are trained for 3,000 steps using an effective batch size of 16. Specifically for the FLUX backbone, we apply a learning rate of $5e-6$, and set the spatial weighting scalar $\alpha = 10$ along with a regularization margin of $\tau = 0.01$ for our alignment objective. Detailed hyperparameter settings for the SDXL backbone are provided in the supplementary material.

6.2 General Results

Quantitative Analysis. Tab. 1 presents a comprehensive quantitative evaluation of our method against standard fine-tuning baselines on HAF-Bench. Notably, our proposed ASAP framework consistently and substantially improves anatomical alignment performance, as indicated by the AER and RSI metrics. At the same time, ASAP maintains—and in many cases surpasses—the performance on other metrics. In contrast, naive DPO leads to significantly worse results on



Fig. 5: Qualitative Comparison with Baselines. Visual comparison against SFT and DPO baselines. While standard SFT fails to resolve complex anatomical artifacts, unbounded DPO reduces these errors but inadvertently distorts background regions with checkerboard artifacts. Conversely, ASAP achieves the precise localized anatomical corrections without compromising the original global visual fidelity.

these metrics. These observations demonstrate that ASAP can achieve improved alignment without sacrificing visual quality or instruction-following ability, our approach substantially reduces the AER across both architectures.

To provide a granular view of these improvements, Fig. 4 illustrates the category-wise AER evaluation across the five HAF-Bench anatomical categories. The tighter polygons for our ASAP-tuned models indicate consistent error reductions across all domains. This balanced improvement across diverse anatomical categories demonstrates the robust generalization of our method, confirming that ASAP effectively reduces anatomical errors regardless of the specific category.

Qualitative Analysis. We visualize the generated samples of models finetuned with different approaches in Fig. 1. The results clearly show that ASAP produces

more anatomically plausible human hands and faces in the generated images. Specifically, while the original FLUX and SDXL base models frequently generate severe anatomical artifacts (*e.g.*, polydactyly, fused fingers and distorted faces), they generally maintain high-fidelity global compositions. ASAP successfully corrects these fine-grained anatomical errors in critical regions like hands and faces, without compromising visual quality and text-image alignment.

To further demonstrate the superiority of our approach over baselines, we provide a qualitative comparison between models fine-tuned with SFT, DPO, and our proposed ASAP framework in Fig. 5. As visually evident, standard SFT struggles to resolve complex anatomical issues, often leaving artifacts like incorrect finger counts or unnatural joint structures largely uncorrected. Meanwhile, although unbounded DPO rectifies certain anatomical errors, it suffers from over-optimization, inadvertently distorting non-targeted background regions. In contrast, ASAP achieves precise anatomical corrections while preserving the global semantic composition and original visual fidelity.

6.3 Ablation Study

Impact of the Regularization Margin

We empirically examine the impact of the regularization margin τ , which governs the trade-off between localized anatomical correction and global semantic preservation, and show the results in Fig. 6. Specifically, when the margin is overly restrictive ($\tau \leq 0.001$), the optimization remains tightly constrained to the base model and thus leads to overly conservative alignment, as shown indicated by the inferior AER scores. Conversely, expanding the margin to excessively large values of $\tau \geq 0.1$ mimics the unbounded objective of standard DPO, resulting in low HPSv3 scores. We empirically observe that $\tau = 0.01$ leads to a Pareto improvements as both metrics improve, compared to the two extreme cases.

6.4 Human Evaluation

To further validate the effectiveness of our proposed method and ensure alignment between automated metrics and human perception, we conducted comprehensive human evaluation on the full HAF-Bench dataset. Specifically, we

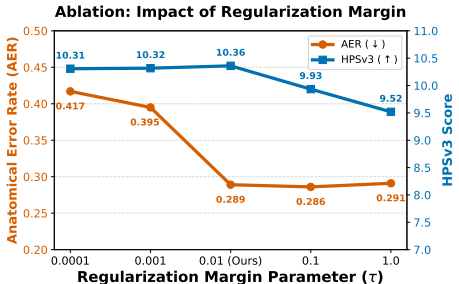


Fig. 6: Impact of the Regularization Margin. The plot illustrates the trade-off between AER $\downarrow$ and HPSv3 $\uparrow$ across different margin parameters. Excessively large margins approximate unbounded DPO, aggressively reducing AER but severely compromising background semantics. Conversely, overly restrictive margins provide insufficient optimization space to correct anatomical artifacts. The optimal margin ($\tau = 0.01$) achieves a critical balance, effectively minimizing anatomical errors while preserving the global generative prior.

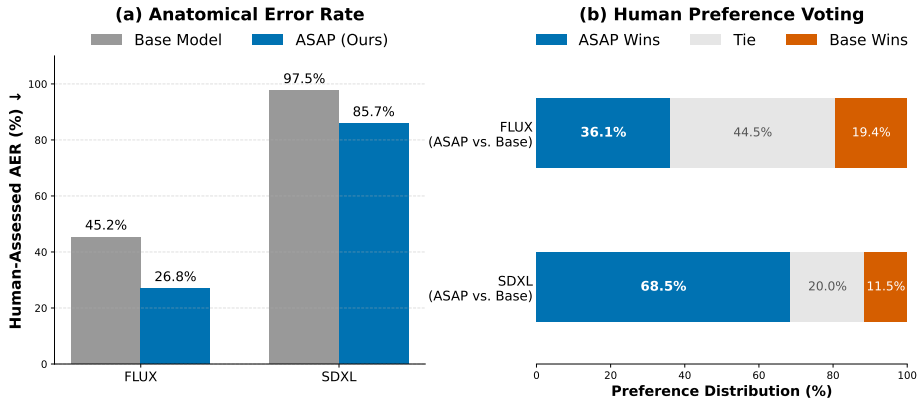


Fig. 7: Human Evaluation. (a) Human Assessed AER. Compared to the original FLUX and SDXL base models, our ASAP-tuned models significantly reduce human-assessed anatomical errors. (b) Human Preference Voting. ASAP achieves a dominant win rate against the base models. This confirms that our method successfully improves anatomical correctness without compromising global image quality.

engaged 11 independent raters who performed binary artifact judgments to determine the human-assessed AER, and selected the more anatomically plausible image from randomized pairs for the RSI metric. As shown in Fig. 7, human feedback consistently corroborates our quantitative findings. Compared to the original base models, ASAP-tuned models significantly reduce anatomical errors, as reflected by substantial reductions in human-assessed AER. This improvement directly translates to human preference voting, where ASAP achieves favorable win rates against both FLUX and SDXL base models. These human evaluation results further validate the effectiveness of our ASAP framework.

6.5 Failure Cases

While our proposed ASAP framework significantly reduces anatomical errors, we observe two primary failure modes, both of which stem from the inherent limitations of the text-to-image foundation models, illustrated in Fig. 8.

Text-Image Misalignment. As illustrated in the left column, when prompts require precise numerical counting (*e.g.*, "counting to five") or specific constraint-based gestures (*e.g.*, "single-hand"), the ASAP-tuned models successfully generate highly realistic and anatomically plausible human images. However, they may occasionally fail to follow the precise semantic constraints of the text. This indicates that while ASAP effectively aligns the generative model toward anatomical correctness, it does not overcome the base model's inherent limitations in numerical reasoning and text-image compositional alignment.

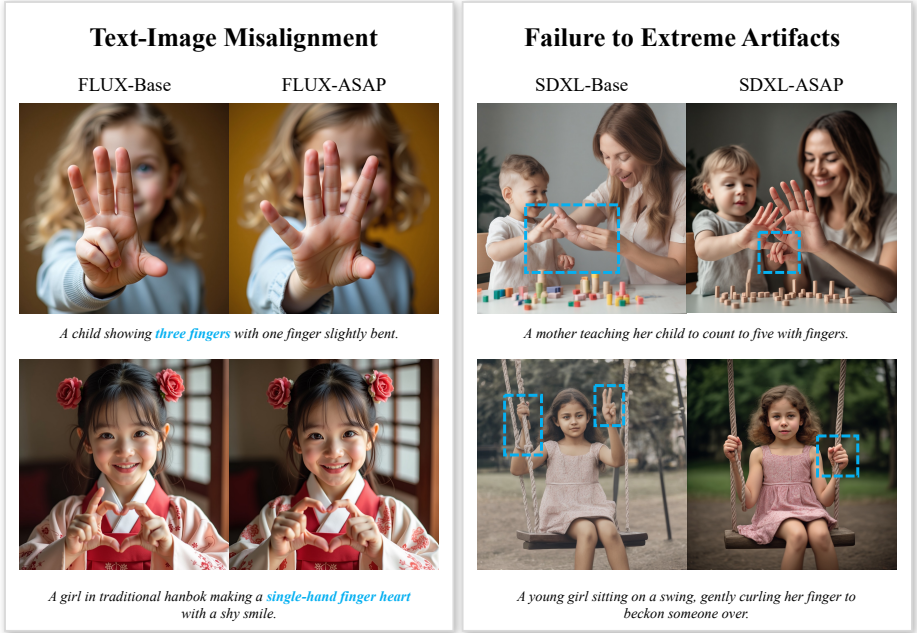


Fig. 8: Failure Case Analysis. (Left) **Text-Image Misalignment:** While the ASAP-tuned models successfully generate anatomically plausible human images, they may occasionally fail to follow precise semantic constraints (e.g., numerical counting or specific gestures). (Right) **Failure to Extreme Artifacts:** In instances where the base model produces severe structural degradation when failing to resolve specific complex details, ASAP still enforces noticeable rectifications (e.g., separating severely fused fingers) but may fall short of achieving perfect anatomical plausibility.

Failure to Extreme Artifacts. We observe that, if the image generated by the base model with some seed contains extreme anatomical artifacts, the finetuned model may still generate artifacts in the corresponding human body parts. For instance, as demonstrated in the right column, when the base model produces severe anatomical artifacts across extensive regions, ASAP still enforces noticeable rectifications (e.g., separating fused fingers), but frequently falls short of achieving perfect anatomical plausibility. This indicates that, as a fine-grained alignment strategy, ASAP fundamentally requires a minimally plausible anatomical foundation from the base generative model.

7 Conclusion

In this paper, we present a preference learning framework that addresses the persistent challenge of anatomical plausibility in human image generation. To provide precise optimization signals, we introduce a localized degradation mechanism that constructs controlled anatomical preference pairs, enabling explicit supervision of common anatomical artifacts. Leveraging these signals, we further

propose a localized, margin-bounded alignment objective that balances targeted anatomical correction with preservation of global semantics.

To systematically evaluate anatomical fidelity, we establish HAF-Bench, a benchmark designed to assess fine-grained anatomical correctness across diverse human-centric scenarios. Extensive experiments demonstrate that our approach significantly reduces anatomical errors across multiple foundation models while maintaining overall visual quality.

Overall, this work bridges the gap between global aesthetic alignment and fine-grained anatomical correctness, offering a principled framework for improving anatomically plausible human image generation.

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